

Notation: The index set for the M subsystems is represented by $\mathcal{I}_M = \{1, 2, \dots, M\}$. The sets of nonnegative and positive integers are denoted by $\mathbb{N}$ and $\mathbb{N}^*$, respectively. The symbol $\doteq$ stands for a first-order approximation, as in [3].

1 Statement of the problem

Consider a switched affine system with dynamics described by a model of the form [1]

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)} \quad (1)$$

where $t \in \mathbb{R}_+$ is the continuous time variable, $x(t) \in \mathbb{R}^n$ is the state vector, $\sigma : \mathbb{R}_+ \rightarrow \mathcal{I}_M$ is a piecewise constant and right-continuous switching function, and $A_j \in \mathbb{R}^{n \times n}$, $b_j \in \mathbb{R}^n$ are the model parameters of the j th subsystem.

Let $x_r(t)$ denote a target state trajectory associated to an admissible switching function $\sigma_r(t)$ of the form

$$\sigma_r(t) = \sigma_r(t_{r_i}), t_{r_i} \leq t < t_{r_{i+1}}, \forall i \in \mathbb{N}^* \quad (2)$$

where t_{r_i} is the i th switching instant. Both $x_r(t)$ and $\sigma_r(t)$ are assumed to be periodic, with period T . The number of switchings within one period is denoted by N .

The problem addressed herein consists in determining perturbations δt_i in the switching instants in order to drive the state from an initial point $x(0)$ to the target trajectory. In what follows, the perturbed switching instants will be denoted by

$$t_i = t_{r_i} + \delta t_i, i \in \mathbb{N}^* \quad (3)$$

The associated switching function $\sigma(t)$ will be defined as

$$\sigma(t) = \sigma_r(t_{r_i}), t_i \leq t < t_{i+1}, \forall i \in \mathbb{N}^* \quad (4)$$

It is assumed that the dwell time at the j th subsystem is lower-bounded by a constant $d_j > 0$, i.e., if $\sigma(t) = j$ and $t_i \leq t \leq t_{i+1}$, then the time interval $t_{i+1} - t_i$ cannot be smaller than d_j .

2 Linearization around the target trajectory

The solution of the state equation (1) at time t_N , starting from the initial condition $x(0)$, is of the form

$$x(t_N) = F_{N-1}F_{N-2} \cdots F_0 x(0) + \left(\sum_{i=0}^{N-2} F_{N-1}F_{N-2} \cdots F_{i+1} g_i \right) + g_{N-1} \quad (5)$$

where

$$F_i = e^{A_i(t_{i+1}-t_i)} \quad (6)$$

$$g_i = \int_0^{t_{i+1}-t_i} e^{A_i \tau} b_i d\tau \quad (7)$$

$$A_i = A_{\sigma(t_i)}, b_i = b_{\sigma(t_i)} \quad (8)$$

In light of (3), (4) and (5) - (8), it can be seen that $x(t_N)$ is a nonlinear function of the switching time perturbations δt_i . In what follows, a linearized model will be derived to describe the relation between $\delta t_1, \delta t_2, \dots, \delta t_N$ and the deviation $x(t_N) - x_r(t_{r_N})$ between the plant state and the target trajectory after N switchings, starting from an initial deviation $x(0) - x_r(0)$.

2.1 Preliminaries

Let $\delta t \in \mathbb{R}^N$ be the following vector of switching time perturbations:

$$\delta t = \begin{bmatrix} \delta t_1 \\ \delta t_2 \\ \vdots \\ \delta t_N \end{bmatrix} \quad (9)$$

The following approximations will be used:

$$e^{At_b} \doteq e^{At_a} + Ae^{At_a}(t_b - t_a) \quad (10)$$

$$\int_{t_a}^{t_b} e^{A\tau} d\tau \doteq e^{At_a}(t_b - t_a) \quad (11)$$

Eq. (10) is a first-order Taylor approximation, whereas (11) corresponds to the left-hand rectangle rule for numerical integration if $t_b > t_a$ and to the right-hand rule otherwise ([2], page 101).

2.2 Main result

Theorem 1: A first order approximation for the relation between the switching-time perturbations and the resulting state deviations is given by

$$x(t_N) - x_r(t_{r_N}) \doteq \Phi [x(0) - x_r(0)] + \Gamma \delta t \quad (12)$$

Proof:

References

- [1] M. Benmiloud, A. Benalia, M. Djemai, and M. Defoort. On the local stabilization of hybrid limit cycles in switched affine systems. *IEEE Transactions on Automatic Control*, 64(2):841–846, 2019.
- [2] C. D. Cantrell. *Modern Mathematical Methods for Physicists and Engineers*. Cambridge University Press, Cambridge, UK, 2000.
- [3] D. E. Kirk. *Optimal Control Theory: An Introduction*. Dover Publications, 2004.